

COSC 594: Interplay Between RL and GenAI

Homework 2

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September 26, 2025

This homework is due on **10/10, 11:59pm**. Write your solution in Latex and submit it on Canvas.

1. (LLM alignment)

- (a) i. (1pt) Recall the policy optimization problem in RLHF,

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right]$$

where $D_{\text{KL}}(\pi(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) := \sum_y \pi(y | x) \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)}$ is the reverse KL divergence between π and π_{ref} . Find a closed-form expression of the optimal policy π^* , and find a closed-form expression of the optimal objective function,

$$\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi^*(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi^*(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right].$$

Solution:

- (b) i. (1pt) Instead of using the reverse KL divergence, we next consider the forward KL divergence,

$$D_{\text{KL}}(\pi_{\text{ref}}(\cdot | x) \| \pi(\cdot | x)) := \sum_y \pi_{\text{ref}}(y | x) \log \frac{\pi_{\text{ref}}(y | x)}{\pi(y | x)}.$$

Using the the forward KL divergence, we define the following policy optimization problem,

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_{\text{ref}}(\cdot | x) \| \pi(\cdot | x)) \right].$$

Find a closed-form expression of the optimal policy π^* , and find a closed-form expression of the optimal objective function,

$$\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi^*(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_{\text{ref}}(\cdot | x) \parallel \pi^*(\cdot | x)) \right].$$

Solution:

- ii. (1pts) Use the expression of π^* in (i) to find a closed-form expression of reward $r(x, y)$ in terms of π^* .

Solution:

- iii. (2pts) Given a preference data set $\mathcal{D} = \{x^{(i)}, y_1^{(i)}, y_0^{(i)}\}_{i=1}^N$, and assume preferring $y_1^{(i)}$ over $y_0^{(i)}$ (i.e., $\mathbb{P}(y_1^{(i)} \succ y_0^{(i)} | x^{(i)})$) follows the Bradley-Terry model with reward r . Rewrite the reward modeling problem,

$$\underset{r}{\text{maximize}} \quad \mathbb{E}_{(x, y_1, y_0) \sim \mathcal{D}} [\log \mathbb{P}(y_1 \succ y_0 | x)]$$

as a preference-based policy optimization problem. If we replace policy π by π_θ with parameter θ , find the policy gradient of the preference-based policy optimization problem.

Solution:

2. (LLM composition)

- (a) i. (1pt) Given K reference models $\{\pi_{\text{ref},k}\}_{k=1}^K$, we consider the policy optimization problem,

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_{\text{ref},k}(\cdot | x)) \right]$$

where $D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_{\text{ref},k}(\cdot | x)) := \sum_y \pi(y | x) \log \frac{\pi(y | x)}{\pi_{\text{ref},k}(y | x)}$ is the reverse KL divergence between π and $\pi_{\text{ref},k}$. Find a closed-form expression of the optimal policy π^* , and find a closed-form expression of the optimal objective function,

$$\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi^*(\cdot | x)} [r(x, y)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(\pi^*(\cdot | x) \parallel \pi_{\text{ref},k}(\cdot | x)) \right].$$

Solution:

- (b) i. (1pt) As we did in LLM alignment, we use the forward KL divergence to define the following policy optimization problem,

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(\pi_{\text{ref}, k}(\cdot | x) \| \pi(\cdot | x)) \right].$$

Find a closed-form expression of the optimal policy π^* , and find a closed-form expression of the optimal objective function,

$$\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi^*(\cdot | x)} [r(x, y)] - \sum_{k=1}^K \beta_k D_{\text{KL}}(\pi_{\text{ref}, k}(\cdot | x) \| \pi^*(\cdot | x)) \right].$$

Solution:

- ii. (1pt) Use the expression of π^* in (i) to find a closed-form expression of reward $r(x, y)$ in terms of π^* .

Solution:

- iii. (2pts) Assume that reward is decomposable,

$$r(x, y) = \sum_{k=1}^K r_k(x, y).$$

For k th reward r_k , we collect a preference data set

$$D_k = \{x^{(i,k)}, y_1^{(i,k)}, y_0^{(i,k)}\}_{i=1}^N$$

and assume preferring $y_1^{(i,k)}$ over $y_0^{(i,k)}$ (i.e., $\mathbb{P}_k(y_1^{(i,k)} \succ y_0^{(i,k)} | x^{(i)})$) follows the Bradley-Terry model with reward r_k . Rewrite the reward modeling problem,

$$\underset{r_1, \dots, r_K}{\text{maximize}} \quad \sum_{k=1}^K \mathbb{E}_{(x^{(k)}, y_1^{(k)}, y_0^{(k)}) \sim D_k} \left[\log \mathbb{P}_k(y_1^{(k)} \succ y_0^{(k)} | x^{(k)}) \right]$$

as a preference-based policy optimization problem. If we replace policy π by π_θ with parameter θ , find the policy gradient of the preference-based policy optimization problem.

Solution: